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MILITARY SPECIFICATION

INSTRUMENTATION ASSEMBLY

(For Exercise Head Mk 85 Mod 0, Mod 1, and Mod 2)

This Specification has been approved by the Naval Ordinance Systems Command, Department of the Navy

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for procurement of three types of instrumentation assemblies.

1.2 Classification. The instrumentation assemblies shall be classed as Type I, Type II, and Type III as specified (see 6.1) and are as follows:

(a) Type I shall be in accordance with DL 2539153.

(b) Type II shall be in accordance with DL 2539071.

(c) Type III shall be in accordance with DL 2539544.

2. APPLICABLE DOCUMENTS

2.1 The following specifications of the issue in effect on the date of invitation for bids form a part of the specifications to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-81059	Transmitter,

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

FSC 1355

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DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

DL 1330423	Instrumentation Assembly, Shipping (Packaging) condition and documents listed thereon
DL 2539153	Instrumentation Assembly Fleet, and documents listed thereon
DL 2539071	Instrumentation Assembly Proofing, and documents listed thereon
DL 2539544	Instrumentation Assembly Fleet and documents listed thereon
NAVORD Drawing 2197301	Chassis, Instrumentation Assembly (Machining)
NAVORD Drawing 2540510	Exercise Head Test Set
NAVORD Drawing 2540558	Instrumentation Test cable
NAVORD Drawing 2540555	Galvo Sim Test Box
NAVORD Drawing 2540565	Patch Plug Exercise Head Test Set

PUBLICATIONS

CD 2539219	Battery Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. Unless otherwise specified in the contract or order, the first article sample shall:

(a) Be manufactured using the procedures proposed for the production lot.

(b) Consist of three instrumentation assemblies.

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(c) Be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the supplier's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Basic requirements

3.2.1 Contractor-furnished data. No data is required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Material and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the instrumentation assembly shall be in accordance with the requirements specified herein and with the applicable documents listed on DL 2539153, DL 2539071, or DL 2539544.

3.3 Performance and product characteristics

NOTE

The numerical values tabulated in Column I of this specification are applicable to the tests conducted in full accordance with quality conformance tests and examinations of 4.7. The values in Column II are applicable to the tests of paragraph 4.5.2.3.2 and Table III test sequence 10. Minimum and maximum values apply to absolute values only (i.e., -40.6 volts direct current (vdc) is greater than -40.1 vdc).

	Column I	Column II
3.3.1 Inputs. Inputs to Type I, Type II, and Type III Instrumentation Assemblies shall require or accept the following inputs:		
3.3.1.1 Type I, Type II, and Type III Instrumentation Assemblies shall be capable of meeting all functional requirements with the following dc inputs:		

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	Column I	Column II
(a) Type I, Type II, and Type III Instrumentation Assemblies		
Voltage:		
+40 VDC	+40 ±2 vdc	+40 ±2 vdc
-40 VDC	-40 ±2 vdc	-40 ±2 vdc
(b) Type II Instrumentation Assemblies		
Voltage:		
-30 VDC LOG AMP	-30 ±2 vdc	-30 ±2 vdc
-10 VDC EXERCISE HEAD	-10.0 ±0.5 vdc	-10.0 ±0.5 vdc

3.3.1.2 Signal inputs. Type I, Type II, and Type III Instrumentation Assemblies shall be capable of operating with the following signal inputs:

(a) Type I, Type II, and Type III Instrumentation Assemblies		
VALID ECHO	0.00 ±0.25 vdc (See NOTE 1)	0.00 ±0.25 vdc
	-14 vdc min (See NOTE 2)	-12 vdc min

[1]NOTE: Indicates the no-voltage condition

[2]NOTE: Indicates the voltage-present or open condition

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	Column I	Column II
RPM (MK 46 MOD 1)	-0.7 ±0.2 vdc at 1850 cps	-0.7 ±0.2 vdc at 1850 cps
RPM IND (MOD 0)	0.0 to +4.0 vdc	0.0 to +4.0 vdc
+40 VDC	+40 ±2 vdc	+40 ±2 vdc
AMPLITUDE THRESHOLD	0.00 ±0.25 vdc (See NOTE 1)	0.00 ±0.25 vdc
	-14 vdc min (See NOTE 2)	-12 vdc min
RANGE GATE	0.00 ±0.25 vdc (See NOTE 1)	0.00 ±0.25 vdc
	-11 vdc min (See NOTE 2)	-10 vdc min
DEPTH INST.	0 to -11 vdc	0 to -11 vdc
+DOPPLER	0.00 ±0.25 vdc (See NOTE 1)	0.00 ±0.25 vdc
	-14 vdc min (See NOTE 2)	-12 vdc min
LEFT COMMAND	0.0 to -7.0 vdc	0.0 to -7.0 vdc
RIGHT COMMAND	0.0 to +7.0 vdc	0.0 to +7.0 vdc
DOWN COMMAND	0.0 to -7.0 vdc	0.0 to -7.0 vdc
UP COMMAND	0.0 to +7.0 vdc	0.0 to +7.0 vdc
-DOPPLER	0.00 ±0.25 vdc (See NOTE 1)	0.00 ±0.25 vdc
	-14 vdc min (See NOTE 2)	-12 vdc min
VALID ECHO INHIBIT	0.00 ±0.25 vdc (See NOTE 1)	0.00 ±0.25 vdc
	-11 vdc min (See NOTE 2)	-10 vdc min

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	Column I	Column II
EXPLODER ARM	-40.0 ±0.1 vdc	-34.0 to -45.0 vdc
+90 VDC INST (SRA)	+90 ±12 vdc	+90 ±12 vdc
+90 VDC INST (INERTIA)	+90 ±12 vdc	+90 ±12 vdc
-40 VDC	-40 ±2 vdc	-40 ±2 vdc
NOISE THRESHOLD	0.00 ±0.25 vdc (See NOTE 1)	0.00 ±0.25 vdc
	-14 vdc min (See NOTE 2)	-12 vdc min
TARGET ANGLE THRESHOLD	0.00 ±0.25 vdc (See NOTE 1)	0.00 ±0.25 vdc
	-12 vdc min (See NOTE 2)	-10 vdc min
INST TIMING MARK	Pulse 5.0 ±1.5 vdc (4 ms) at 0.50 ±0.01 sec intervals	Pulse 5.0 ±1.5 vdc at 0.50 ±0.01 sec intervals
PING DRIVE	0.00 ±0.25 vdc (See NOTE 1)	0.00 ±0.25 vdc
	-11 vdc min (See NOTE 2)	-10 vdc min
(b) Type II Instrumentation Assembly		
FREQ DISCR OUT	-10.0 to +10.0 vdc	-10.0 to +10.0 vdc
INST ROLL DRIVE	-40.0 to -2.0 vdc	-40.0 to -2.0 vdc
CHAMBER PRESS.	0 to -11 vdc	0 to -11 vdc
AGC THRESHOLD	-18 ±4 vdc	-18 ±4 vdc
INST PITCH PEND	0.0 to 3.5 vrms (See NOTE 3)	0.0 to 3.5 vrms
INST ROLL PEND	0.0 to 3.5 vrms (See NOTE 3)	0.0 to 3.5 vrms

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	Column I	Column II
LEFT CHANNEL INPUT SOUND	f_o ±300 cps (See NOTE 4)	f_o ±300 cps
PITCH ANGLE DATA	-12.0 to +12.0 vdc	-12.0 to +12.0 vdc
FREQ INDICATOR	-0.5 to +4.0 vdc	-0.5 to +4.0 vdc
INST YAW DRIVE	-40.0 to -2.0 vdc	-40.0 to -2.0 vdc
INST YAW RATE GYRO	0.0 to 3.5 vrms (See NOTE 3)	0.0 to 3.5 vrms
SAMPLE	-11 vdc min	-11 vdc min
YAW ANGLE DATA	-12.0 to +12.0 vdc	-12.0 to +12.0 vdc
INST PITCH DRIVE	-40.0 to -2.0 vdc	-40.0 to -2.0 vdc
INST PITCH RATE GYRO	0.0 to 3.5 vrms (See NOTE 3)	0.0 to 3.5 vrms
TIME SHARE TRIGGER	Pulse 5.5 ±1.0 volts (2 ms)	Pulse 5.5 ±1.0 volts (2 ms)
(c) Type II alternate inputs available		
PHASE DETECTOR OUTPUT	-6 to +6 vdc	-6 to +6 vdc
AFT INST SPARE	open	open
AC INST NO. 1	open	open
AC INST NO. 2	open	open
ELEVATOR INST	0 to -11 vdc	0 to -11 vdc

3NOTE: In-phase and 180° out-of-phase with 26 vrms 4.8 kc reference.

4NOTE: See MIL-T-81059 for f_o

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	Column I	Column II
UPPER RUDDER INST	0 to -11 vdc	0 to -11 vdc
LOWER RUDDER INST	0 to -11 vdc	0 to -11 vdc
INST ROLL RATE GYRO	0.0 to 3.5 vrms (See NOTE 3)	0.0 to 3.5 vrms
(d) Type I Instrumentation Assemblies		
COURSE GYRO DEMOD OUT	±1.0 vdc max	±1.0 vdc max
(e) Type II and Type III Instrumentation Assemblies		
C. G. IN-NORM terminals jumpered: COURSE GYRO DEMOD OUT	±1.0 vdc max	±1.0 vdc max
C. G. IN-HATS terminals jumpered: COURSE GYRO IN	0.000 to +0.385 vdc max	0.000 to +0.385 vdc max

3.3.2 Outputs. Signal outputs from Type I, Type II, and Type III Instrumentation Assemblies shall be:

(a) Type I, Type II, and Type III Instrumentation Assemblies		
3 MIN SNAKE	0.0 ±0.2 vdc (See NOTE 1)	0.0 ±0.2 vdc
	open (See NOTE 2)	open
ALD/HLD RESET	-5.0 vdc min (See NOTES 1 and 5)	-4.0 vdc min
	open (See NOTE 2)	open

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	Column I	Column II
DCO SW/TERM SQUIB RET	-5.0 vdc min (See NOTE 5)	-4.0 vdc min
DCO SWITCH	-5.0 vdc min (See NOTE 5)	-4.0 vdc min
DCO SQUIB	-2.5 vdc min (See NOTE 5)	-2.0 vdc min
STEERAWAY DISABLE	0.0 ±0.2 vdc (See NOTE 1)	0.0 ±0.2 vdc
	open (See NOTE 2)	open
(b) Type II Instrumentation Assembly		
PITCH RATE GYRO demodulator output	-1.2 to +1.2 vdc	-1.2 to +1.2 vdc
YAW RATE GYRO demodulator output	-1.2 to +1.2 vdc	-1.2 to +1.2 vdc
PITCH PEND demodulator output	-1.2 to +1.2 vdc	-1.2 to +1.2 vdc
ROLL PEND demodulator output	-1.2 to +1.2 vdc	-1.2 to +1.2 vdc
Log amplifier output	+0.4 to +6.0 mvdc	+0.4 to +6.0 mvdc
(c) Type II Alternate outputs available		
ROLL RATE GYRO demodulator output	-1.2 to +1.2 vdc	-1.2 to +1.2 vdc

5NOTE: Referenced to BATTERY +5.4 V. When delivering 160 milliamperes (ma) minimum.

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3.3.3 Performance characteristics. Type I, Type II, and Type III Instrumentation Assemblies shall conform to the following characteristics.

3.3.3.1 Type I and Type II Instrumentation Assemblies shall:

(a) Provide resistive and capacitive circuitry for signal conditioning.

(b) Consist of a battery pack comprising two 2.7 volt batteries connected in series.

(c) Consist of an ALD/HLD RESET switch to provide -5.4 volts to ADL/HLD RESET output.

(d) Provides a circuit with provisions for grounding PING DRIVE, STEERAWAY DISABLE, and for completing the circuit of the 3-Minute timer.

(e) Consist of a 3-Minute timer circuit to provide a ground to the 3 MIN SNAKE output, three minutes after power is applied.

3.3.3.2 Type II Instrumentation Assembly shall:

(a) Provide additional resistive and capacitive circuitry for signal conditioning.

(b) Consist of a logarithmic amplifier to convert the amplifier modulation of the LEFT CHANNEL INPUT SOUND signal to a direct current signal, logarithmically proportional to the input modulation.

(c) Consist of five demodulator plug-in component board assemblies to convert the amplitude modulation of 4.8 ±1.2 kc input signals to a variable voltage, direct-current signal. The output shall be negative with respect to common when the input carrier is in phase within 10 degrees of the reference voltage, and positive with respect to common when the carrier is 180 ±10 degrees out of phase with the reference voltage.

(d) Consist of a time share multivibrator plug-in component board assembly to provide a time sequence of 0.5 seconds rate for switching PITCH PEND and ROLL PEND demodulator output signals.

3.3.3.3 Type II and Type III Instrumentation Assemblies shall:

(a) Provide a circuit for selecting either of two types of course gyroscope inputs to the oscillograph as follows:

(1) With terminals C. G. IN and NORM jumpered together, the instrumentation assembly will process the COURSE GYRO DEMOD OUT input from the demodulator.

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(2) With terminals C. G. IN and HATS jumpered together, the instrumentation assembly will process the course gyroscope input from COURSE GYRO IN.

3.3.4 Life. Life requirements for unit conditions (see 6.3.1) shall be as follows:

3.3.4.1 Operating. The instrumentation assembly shall meet requirements of 3.3.1 through 3.3.3 operation throughout its mission time of 9 minutes.

3.3.4.2 Total operating life. The exercise head shall be capable of satisfactory operation for a minimum of 200 cycles of operation as defined herein and for a period of 200 hours continuous operation with the application of steady state power. A cycle of operation shall consist of a period of nine minutes operation with a cooling off period of 10 minutes preceding each cycle. The life requirement shall begin at the date of acceptance by the procuring agency. Individual items with lesser life expectancy may be replaced or disabled upon conclusion of their respective life.

3.4 Environmental requirements. Unless otherwise specified, the instrumentation assembly shall be capable of conforming to performance and product characteristics requirements of this specification after being exposed, at any temperature in the applicable range, to the environments under all conditions specified in this section.

3.4.1 Vibration. The instrumentation assembly shall be capable of withstanding vibration shown in Table I.

Table I

Vibration Schedule

Condition (See 6.3.1)	Double Amplitude (in) minimum	Acceleration (g's) minimum	Frequency (cps)
Nonoperating	---	1.0	5 to 60
Ready nonoperating, Ready	0.2	---	5 to 17
	---	3.0	17 to 500

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3.4.2 Shock. The instrumentation assembly packaged in accordance with DL 1330423 (see 5.1.1) shall be capable of withstanding the shock of being dropped on a nonyielding surface from a height of 36 inches. Additionally, the instrumentation assembly, in the condition shown in Table II, shall withstand shocks specified in Table II.

Table II

Shock Schedule

Condition (See 6.3.1)	Peak Magnitude (g's)	Duration (milliseconds)	Acceleration Direction*	Number of Impacts
Ready non-operating	60 ±10%	8.0 to 10.0	3 fwd or aft axis 3 left or right axis 3 up or down axis	9
Ready	45 ±10%	18.0 to 20.0	Long. fwd axis	3
	85 ±10%	0.5 to 1.5	3 up or down axis 3 right or left axis	6
	300 ±10%	0.5 to 1.5	Long. aft axis	3

*See Figure 1 for axes of instrumentation assembly.

3.4.3 Temperature. The instrumentation assembly shall be capable of withstanding temperatures for the following specified conditions:

3.4.3.1 Nonoperating and ready nonoperating. The instrumentation assembly shall be capable of withstanding ambient temperatures between -45°F and 140°F wherein the duration of the upper temperature does not exceed four hours at any one time, and the temperature is stabilized in a temperature range of 60°F to 95°F prior to any performance test.

3.4.3.2 Ready and operating. The instrumentation assembly, when operated in an environment with a temperature between 32°F and 98°F, shall meet the performance and product characteristics of this specification

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(Column II values) after exposure to temperatures between 0°F and 140°F, for a period wherein the durations of the upper temperature limit do not exceed 4 hours at any one time.

3.4.4 Pressure. The instrumentation assembly shall be capable of withstanding external pressures of from 2.7 to 30.0 psia.

3.4.5 Humidity. The instrumentation assembly shall withstand exposure to 95 percent relative humidity at +110 degrees F for eight hours without degradation of finish or functional capabilities.

3.5 Workmanship. Workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with requirements of this specification. All parts used in the instrumentation assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. Standards of workmanship exhibited in any approved first article sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the instrumentation assembly shall be classified as follows:

(a) First article inspection

(1) First article sample (see 4.3.1)

(2) First article tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

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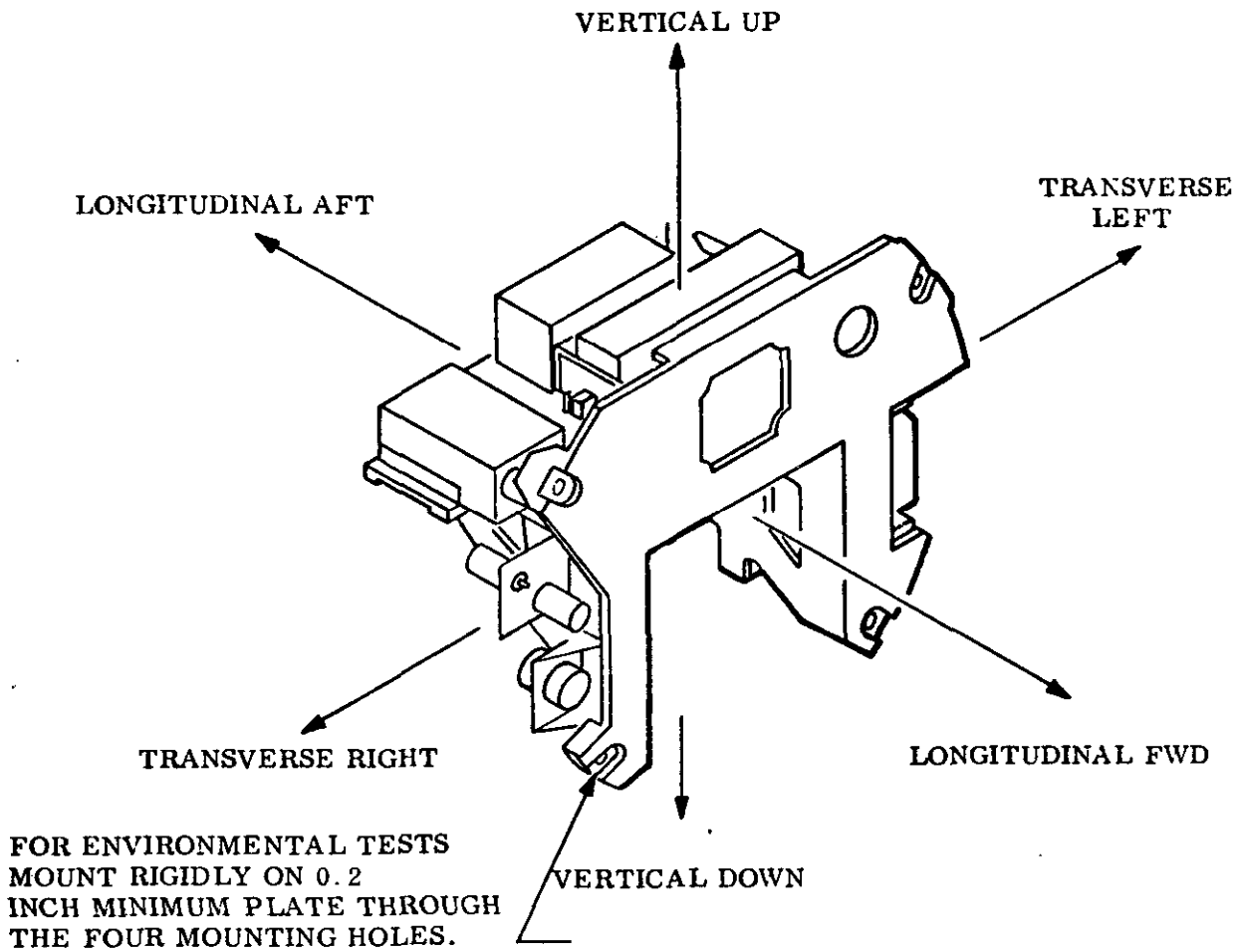


Figure 1. Axes of Instrumentation Assembly

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(c) Quality conformance inspection

(1) Quality conformance sampling (see 4.3.3)

(2) Quality conformance tests and examinations (see 4.7)

4.3 Sampling

4.3.1 First article sample. A first article sample of three instrumentation assemblies, manufactured in accordance with 3.1, shall be subjected to first article tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the first article sample shall be based on no defects in the sample. Further production of the instrumentation assembly by the subsupplier, prior to approval of the first article sample, shall be at supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from units having satisfactorily passed quality conformance tests and examinations of 4.7. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

(a) Step 1. Upon initiation of production randomly select 1 unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2. Each sample so selected shall be numbered consecutively starting with Sample Number 1.

(b) Step 2. Randomly select 1 unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively starting with a sample number one greater than the highest sample number selected in Step 1.

4.3.2.1 Defective units. When a defective unit (see 6.3.8) is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

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(d) Submit to the procuring activity for approval the proposed re-test program to verify that the corrective action eliminated the defect.

(e) Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Quality conformance sampling. Tests and examinations specified in 4.7 shall be conducted on each instrumentation assembly to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the instrumentation assembly shall be subjected to tests under the following conditions:

(a)	Temperature	Room ambient (60 to 95 degrees F)
(b)	Humidity (relative)	Room ambient, up to 95 percent
(c)	Altitude	Normal ground atmospheric pressure
(d)	Vibrations	None

4.5 First article tests. The instrumentation assembly within first article sample (identified as Instrumentation Assemblies 1, 2, and 3) shall be subjected to tests specified in Table III. Certified inspection data covering results of first article tests shall be forwarded to the Naval Ordnance Systems Command. Certified inspection data shall consist of the following:

(a) Data of tests and names of persons conducting tests.

(b) Run or test plan used.

(c) Serial number and type of instrumentation assembly.

(d) Test results of first article tests (4.5.2 through 4.5.3).

Table III

First Article Test Schedule

Test Sequence	Test Description	Instrumentation Assembly Number		
		1	2	3
1	Quality conformance tests and examinations (4.7).	x	x	x
2	Vibration (4.5.2.1).	x		x

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Table III (Cont'd)

First Article Test Schedule

Test Sequence	Test Description	Instrumentation Assembly Number		
		1	2	3
3	Shock (4.5.2.2).	x	x	
4	Temperature, storage (4.5.2.3.1).	x	x	
5	Temperature, operating (4.5.2.3.2).	x	x	
6	Pressure test (4.5.2.4).	x	x	
7	Humidity (4.5.2.5).	x		x
8	Life test (4.5.2.6).	x		
9	Quality conformance tests and examinations (4.7).		x	x
10	Quality conformance tests and examinations (4.7) with Column II values specified in 3.3	x		

4.5.1 First article test equipment. The following items of test equipment are required to perform first article tests specified herein:

4.5.1.1 A vibration table capable of vibrating the instrumentation assembly, plus fixture weight at 3 g's maximum over a frequency range of 5 to 500 cycles per second with a maximum double amplitude of 0.2 inch.

4.5.1.2 A shock testing machine capable of applying shocks of up to 300 g's maximum at durations from 0.5 to 20 milliseconds.

4.5.1.3 A temperature chamber capable of programmed temperature variations over a range of -45 to +140 degrees F.

4.5.1.4 A humidity chamber capable of providing a constant humidity condition of 95 percent at 100 degrees F for extended periods up to 8 hours.

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4.5.1.5 Items of test equipment specified in 4.7.

4.5.2 Detailed first article tests

4.5.2.1 Vibration

4.5.2.1.1 Nonoperating. The packaged instrumentation assembly (see 5.1.1) shall be attached to a rigid support fixture oriented with respect to the three orthogonal axes of the shipping container, one axis being along the cylindrical axis of the container. The resonance of the packaged instrumentation assembly in each of the three axes shall be determined by varying the frequency of the applied vibration slowly through the specified range shown in Table I. The packaged instrumentation assembly shall then be vibrated at the most severe resonance for a period of one hour for each of the three orthogonal axes of the shipping container.

4.5.2.1.2 Ready nonoperating and ready. The unpackaged instrumentation assembly mounted on a support frame, shall be rigidly attached to the vibration equipment. This support frame shall be capable of transmitting the vibrations listed in Table I to the instrumentation assembly along the axes shown in Figure 1. Vibrations of Table I shall be accomplished by cycling, the frequency rate being logarithmic with 15 minutes required to proceed from the lower to higher and back to the lower extremity of the frequency range. The total vibration time in each axis shall be 180 minutes (12 cycles), in a room ambient condition.

4.5.2.2 Shock

4.5.2.2.1 Nonoperating. The packaged instrumentation assembly (see 5.1) shall be subjected to free fall drop tests on corners and surfaces of the container. The packaged unit shall be dropped from a height of 36 inches above an impact surface which shall be essentially nonyielding stone, concrete or similar material. The container, with the packaged unit inside, shall be dropped in sequence on corners 5-1-2, 6-3-4, 5-2-3, and 6-1-4, and side 1 and face 5 as defined in Figure 2 on the following page.

4.5.2.2.2 Ready nonoperating and ready. Mount the unpackaged instrumentation assembly onto a shock test fixture so that a minimum of 90 percent of table input of the fixture will be applied to the instrumentation assembly under test. Subject the instrumentation assembly to shocks listed in Table II, in the conditions specified. Duration shall be measured by an accelerometer mounted on input table of fixture to indicate applied shock.

4.5.2.3 Temperature

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4.5.2.3.1 Temperature, nonoperating. The instrumentation assembly preserved and packaged in accordance with 5.1.1 shall be placed in a temperature controlled chamber. The chamber shall then be subjected to the following temperature cycles:

(a) Heat chamber to +140 +5 -0 degrees F in 2.5 ±0.5 hours.

(b) Maintain at +140 +5 -0 degrees F for 4 hours minimum, then within 2.5 ±0.5 hours cool chamber to -45 +0 -5 degrees F.

(c) Maintain at -45 +0 -5 degrees F for 4 hours minimum, then within 2.5 ±0.5 hours return chamber to room temperature.

Repeat this schedule for a minimum of three cycles.

4.5.2.3.2 Temperature, operating. The instrumentation assembly, mounted as described in Figure 1, shall meet quality conformance test requirements of 4.7 except as specified below.

(a) The instrumentation assembly shall be stabilized for one hour at a temperature of +140 +5, -0 degrees F, then transferred to an environment at +98 +5, -0 degrees F and tested to assure conformance with performance and product characteristics of this specification (Column II values). This conformance shall be shown to exist for any combination of power supply and input voltages tabulated in 3.3.1.1 (Column II) and 3.3.1.2 (Column II) of this specification.

(b) The instrumentation assembly shall be stabilized for one hour at a temperature of zero -5, +0 degrees F, then transferred to an environment at +32 -3, +0 degrees F and tested to assure conformance with performance and product characteristics of this specification (Column II values). This conformance shall be shown to exist for any combination of power supply and input voltages tabulated in 3.3.1.1 and 3.3.1.2 of this specification.

4.5.2.4 Pressure. Subject the unpackaged instrumentation assembly to pressures of 2.7 psia maximum and 30.0 psia minimum for 30 minutes at each pressure extreme.

4.5.2.5 Humidity. The unpackaged instrumentation assembly shall be placed in a suitable test chamber and be subjected to at least 95 percent relative humidity at a temperature of 110°F +5°F -0°F for a period of 8 hours minimum. Upon completion of this test, remove the instrumentation assembly from the chamber and immediately initiate the quality conformance test (para 4.7).

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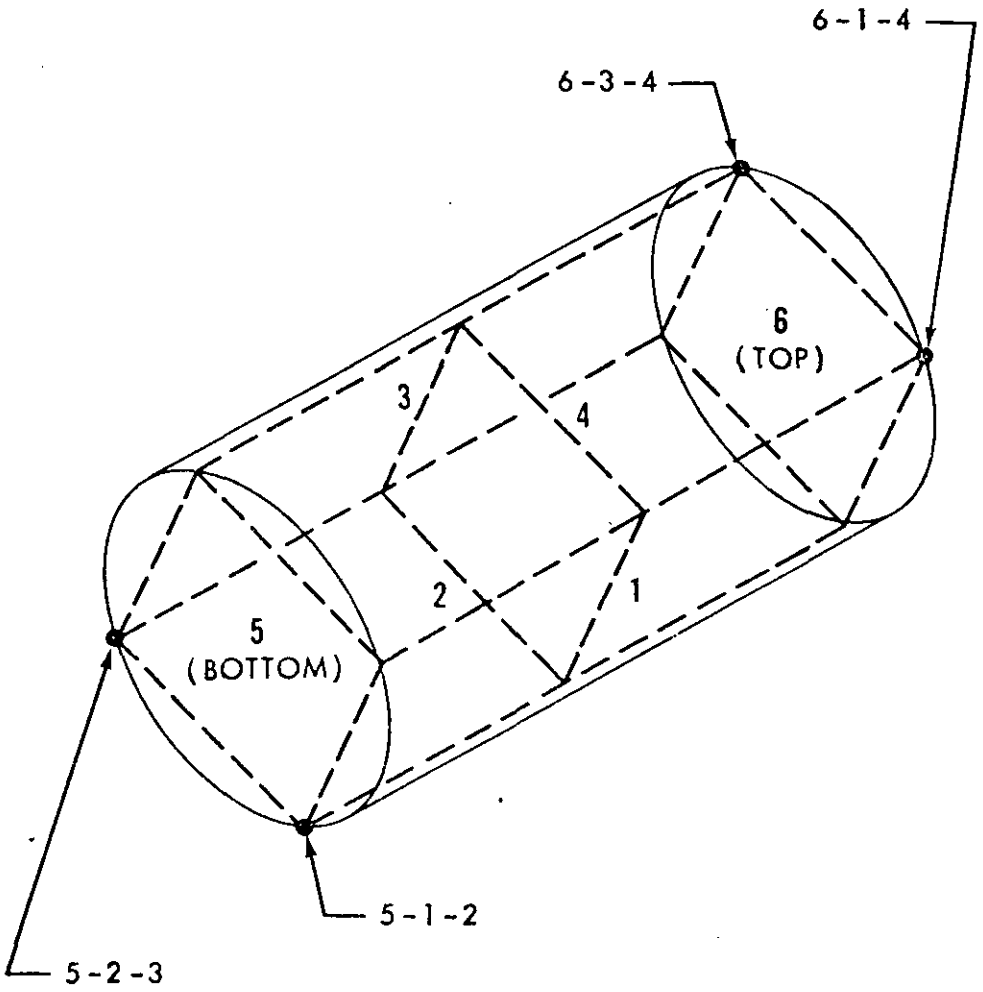


Figure 2. Definition of Faces and Corner of Shipping Container DL 1330423

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4.5.2.6 Life test

4.5.2.6.1 Cycling test. The instrumentation assembly shall be energized for 200 cycles wherein the duty cycle shall be nine minutes on and ten minutes off. On each cycle all functions shall be properly loaded and energized.

4.5.2.6.2 Steady state test. The instrumentation assembly shall be operated for a period of 200 continuous hours with steady state power applied.

4.5.3 First article acceptance criteria. Acceptance shall be based on satisfactory completion of test sequence item 9 or 10 on Table III as applicable.

4.6 Periodic production test. Periodic production sample, selected in accordance with 4.3.2, shall be subjected to the tests of 4.5.

4.6.1 Periodic production test equipment. The items of test equipment shall be those specified for the conduction of all tests of Sections 4.5 and 4.7.

4.6.2 Periodic production test schedule. The periodic test sample shall be subjected to the tests noted in Table IV in the order specified.

Table IV

Periodic Production Test Schedule

Test	Sample and Test Sequence		
	Sample No. 1, 4, Etc.	Sample No. 2, 5, Etc.	Sample No. 3, 6, Etc.
Nonoperating vibration (4.5.2.1)	1*		1*
Shock (4.5.2.2)		1*	3*
Temperature Nonoperating (4.5.2.3.1)		2	4
Hi-Temperature Operating (4.5.2.3.2,a)		3	5
Lo-Temperature Operating (4.5.2.3.2,b)	3		2

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Table IV (Cont'd)

Periodic Production Test Schedule

Test	Sample and Test Sequence		
	Sample No. 1, 4, Etc.	Sample No. 2, 5, Etc.	Sample No. 3, 6, Etc.
Pressure (4.5.2.4)	2		6
Humidity (4.5.2.5)			7
Life (4.5.2.6.1)			9
Quality conformance (Column I) (4.7)			8
Quality conformance (Column II) (4.7)			10

*Indicates visual examination after test.

4.6.3 Periodic production acceptance criteria. Acceptance criteria shall be in accordance with 4.3.2.1 and 4.7.1.3.

4.7 Quality conformance tests and examinations. Unless otherwise specified, each instrumentation assembly delivered under contract shall be subjected to the tests and examinations specified herein. The test procedures are basically for a Type I configuration. Test procedures for a Type II configuration are the same as for Type I except for those additional tests specified. Indicate on the Report Form the type of configuration being tested.

4.7.1 Conditions

4.7.1.1 Test sequence. Unless otherwise noted, sequence of testing procedures presented herein is mandatory and must be performed in order shown.

4.7.1.2 Test results. Where space is provided in Appendix Report Form to indicate value or specific reading obtained, the specific reading shall be recorded if a limit or limits are given in Limits column for the test. If limits are not stated in Limits column, the individual performing test or inspection shall initial in spaces provided in Reading column, indicating that requirements were satisfactorily met. A copy of the report form, bearing recorded data, shall be shipped with the inspected unit.

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4.7.1.3 Acceptance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the Naval Ordnance Systems Command via the cognizant Government representative.

4.7.2 Quality conformance test equipment. Items of test equipment shown in Table V are required to perform quality conformance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals. Procedures for and results of all calibrations shall be subjected to approval of the cognizant Government inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration.

Table V

Quality Conformance Test Equipment

Item	Identification
Exercise Head Test Set	NAVORD Dwg 2540510
Galvo Sim Test Box	NAVORD Dwg 2540555
Instrumentation Test Cable	NAVORD Dwg 2540558
Patch Plug PT 4810	NAVORD Dwg 2540565
Digital DC Voltmeter (0.001 to 999 vdc; ±0.2% of reading, ±1 count accuracy)	Hewlett-Packard Model 405 BR
AC VTVM (Range: 1.0 mv to 300 vrms: Accuracy: ±1% 50 cps to 500 kc/s. Frequency Range: 10 cps to 4 mc/s.)	Hewlett-Packard Model 400 H
Multimeter (Resistance: 0 to 20 megohms)	Simpson, Model 260

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4.7.3 Test procedure

4.7.3.1 Continuity tests. Connect an ohmmeter between the following connector pins. The ohmmeter shall indicate 0.5 ohm maximum.

From	To
J4806-A	J4806-a
J4806-K	J4805-V, T
J4806-K	J4804-s, t
J4806-K	J4810-X
J4806-S (See NOTE 6)	J4810-C
J4806-T	J4810-F
J4806-U	J4810-E
J4806-X	J4810-KK
J4806-Y	J4810-JJ
J4806-b	J4810-AA
J4806-c	J4810-CC
J4806-c	J4805-K
J4806-d	J4805-H
J4806-e	J4805-F
J4806-h	J4805-W
J4806-j	J4805-B
J4806-m	J4805-L
J4806-u	J4810-A
J4806-w	J4810-B

6NOTE: For Type I only.

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From	To
J4806-x	J4810-p
J4806-y	J4805-M
J4806-AA	J4805-Z
J4806-BB	J4805-Y
J4806-CC	J4810-N
J4806-DD	J4810-R
J4806-EE	J4810-H
J4806-EE	J4804-Z
J4806-EE	J4804-b
J4806-EE	J4804-d
J4806-EE	J4804-f
J4806-EE	J4804-h
J4806-EE	J4804-j
J4806-EE	J4804-m
J4806-EE	J4804-p
J4806-EE	J4803-V
J4806-FF	J4805-N
J4805-A	J4810-MM
J4805-C	P2302-A
J4805-D	J4810-EE
J4805-E	J4810-DD
J4805-J	J4810-FF
J4805-U	J4810-LL

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From	To
J4805-AA	J4810-K
J4805-BB	J4810-S
J4805-CC	J4804-q
J4805-DD	J4804-r
J4804-s, t	TB4802, GND
J4803-L	TB4802, PING DRIVE
J4810-L	J4803-D
J4810-M	TB4802, STEERAWAY DISABLE
J4810-V	J4804-X
J4806-S (See NOTE 7)	TB4802, C. G. IN
J4810-J (See NOTE 7)	TB4802, HATS
J4810-C (See NOTE 7)	TB4802, NORM

4.7.3.2 Type II continuity and resistance tests

4.7.3.2.1 Remove connector P4808 and P4809 from logarithmic amplifier. Connect an ohmmeter between the following connector pins. Ohmmeter shall indicate 0.5 ohm maximum.

From	To
P4808-A	P2302-M
P4808-B	P2302-P
P4808-C	P2302-N
P4809-A	J4804-U
P4809-B	J4804-V

7NOTE: For Type II and Type III only.

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From	To
J4806-EE	J4804-K
J4806-EE	J4804-M
J4806-EE	J4804-R
J4806-EE	J4804-S

4.7.3.2.2 Connect ohmmeter between connectors P4809-P and J4806-z. Ohmmeter shall indicate 313 to 347 ohms.

4.7.3.2.3 Reconnect connectors P4808 and P4809 to logarithmic amplifier.

4.7.3.3 Resistance tests

4.7.3.3.1 Connect ohmmeter between connectors J4804-Y and J4810-K. Ohmmeter shall indicate 256 k to 284 k ohms.

4.7.3.3.2 Connect ohmmeter between connectors J4804-i and J4810-S. Ohmmeter shall indicate 256 k to 284 k ohms.

4.7.3.3.3 Connect ohmmeter between connectors J4806-J and J4804-g. Ohmmeter shall indicate 31.3 k to 34.7 k ohms.

4.7.3.3.4 Connect ohmmeter between connectors J4810-D and J4804-k. Ohmmeter shall indicate 950 to 1050 ohms.

4.7.3.3.5 Connect ohmmeter between connectors J4810-J and J4804-n. Ohmmeter shall indicate 2.56 to 2.84 k ohms.

4.7.3.4 ALD/HLD RESET switch test

4.7.3.4.1 Connect an ohmmeter between pins HH and CC of connector J4810.

4.7.3.4.2 Depress ALD/HLD RESET switch on instrumentation assembly and observe ohmmeter. Ohmmeter shall indicate 0.2 ohm maximum when switch is depressed and an open circuit when switch is released.

4.7.3.5 Initial test setup

4.7.3.5.1 Verify that main POWER circuit breaker on test set is OFF.

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4.7.3.5.2 Connect instrumentation assembly to test set as shown in Figure 3.

4.7.3.5.3 Position exercise head test panel controls to the following positions:

Control	Position
GALVO switches	center
DC INPUT CALIBRATE control	500
AC INPUT CALIBRATE control	500
OSCILLOGRAPH MOTOR switch	down
OSCILLOGRAPH LAMP switch	down
EXERCISE HEAD PWR switch	down
RANGE switch	R > B
LOGIC POWER switch	OFF
DCO-ALD RESET switch	center
TURNAWAY KO switch	up
DC VOLTS switch	OFF
INPUT SOUND LEVEL switch	-40
AC SELECT switch	DEMOD
RPM FREQ control	2000

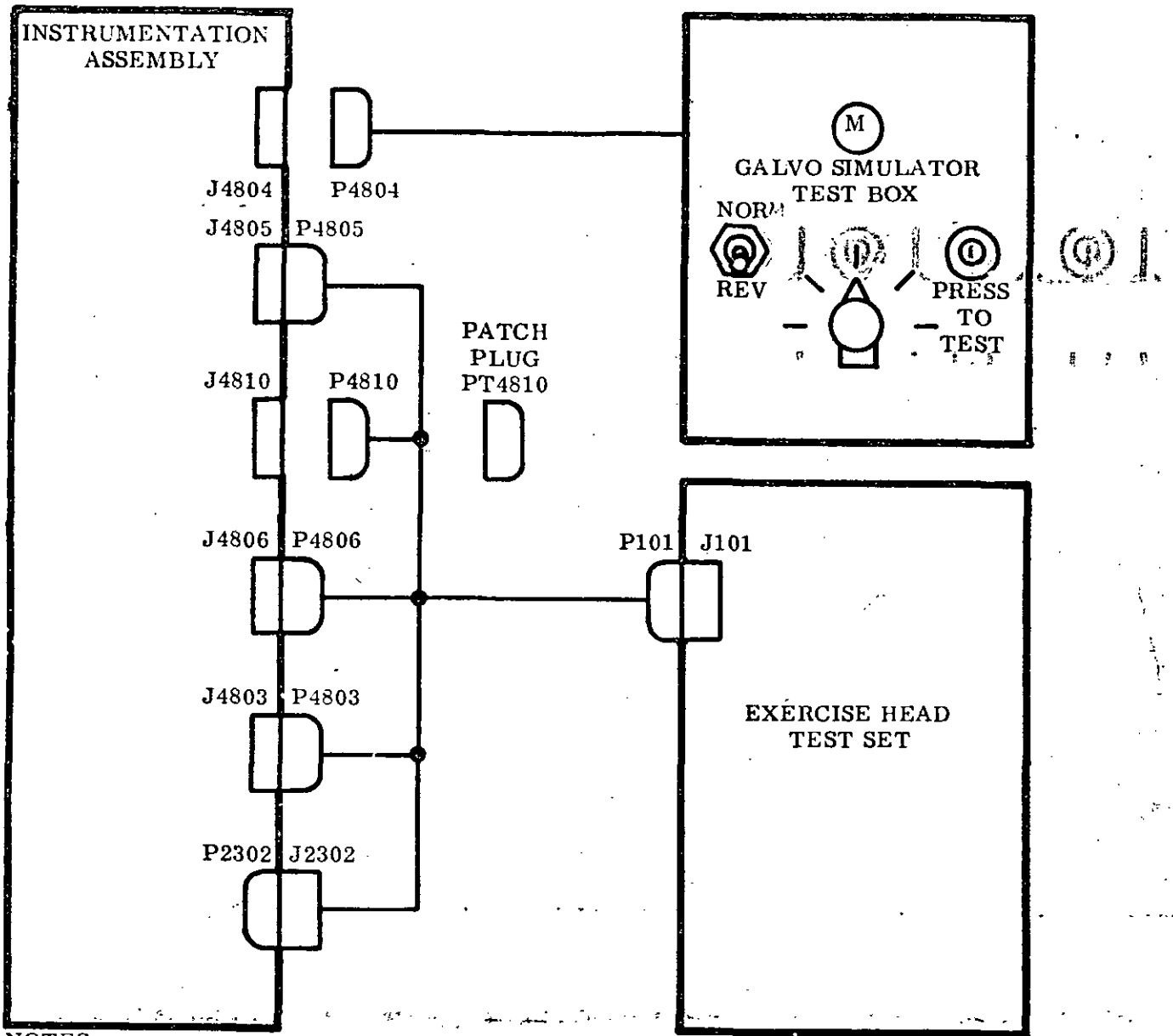
4.7.3.5.4 Position main POWER circuit breaker to ON and connect digital dc voltmeter to meter MONITOR jacks on test panel.

4.7.3.5.5 Position DC VOLTS switch on test panel to +40 and adjust +40V power supply for +40.0 ±0.5 vdc indication on digital DC voltmeter.

4.7.3.5.6 Position DC VOLTS switch on test panel to -40 and adjust -40V power supply for -40.0 ±0.5 vdc indication on digital DC voltmeter.

4.7.3.6 3-minute snake

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NOTES:
1. P4810 (SEE 4.7.3.7.1)
2. P4804 (SEE 4.7.3.8.1)
3. PT4810 (SEE 4.7.3.8.22)

Figure 3. Instrumentation Assembly Test Setup

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4.7.3.6.1 Temporarily connect jumper between 3 MIN TIMER and GND pins on junction board TB4802 of instrumentation assembly.

4.7.3.6.2 Place EXERCISE HEAD PWR switch to up (on) position and simultaneously record MINUTES meter indication.

4.7.3.6.3 Record MINUTES meter indication when 3 MIN SNAKE indicator illuminates.

4.7.3.6.4 The difference between MINUTES meter indication in 4.7.3.6.2 and MINUTES meter indication in 4.7.3.6.3 shall be 2.5 to 3.5 minutes.

4.7.3.7 Battery load test

CAUTION

During Battery Load test, hold LOAD switch on test panel only long enough to observe a panel meter indication.

4.7.3.7.1 Connect P4810 to J4810. (See Figure 3.) DCO SQUIB and TERMINATION SQUIB indicators shall illuminate.

4.7.3.7.2 Position DC VOLTS switch to BT1-2.7 and momentarily hold LOAD switch to BT1 position. Panel meters shall indicate -2.5 vdc minimum.

4.7.3.7.3 Position DC VOLTS switch to BT2-2.7 and momentarily hold LOAD switch to BT2. Panel meter shall indicate -2.5 vdc minimum.

4.7.3.7.4 Disconnect P4810 from J4810. DCO SQUIB and TERMINATION SQUIB indicators shall be extinguished.

4.7.3.8 Signal conditioning circuits

4.7.3.8.1 Connect GALVO SIM test box to instrumentation assembly. (See Figure 3.)

NOTE

DC INPUT CALIBRATE dial shall be set below 500 for a negative voltage and above 500 for a positive voltage. AC INPUT CALIBRATE dial shall be set above 500 for a 0° ϕ rms voltage and below 500 for a 180° ϕ rms voltage. Position GALVO SIM polarity switch to NORM. Depress PRESS TO TEST button on GALVO SIM to obtain meter indications.

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4.7.3.8.2 Position switch on GALVO SIM test box to 1 and connect digital DC voltmeter to DC INPUT CALIBRATE jacks on test panel. Adjust DC INPUT CALIBRATE control for +25.0 ±0.1 vdc indication on digital dc voltmeter. Position VALID ECHO switch on test panel to down. GALVO SIM meter shall indicate 36 to 44 microamperes.

4.7.3.8.3 Return VALID ECHO switch to center and position GALVO SIM switch to 2. Adjust DC INPUT CALIBRATE control for +0.40 ±0.01 vdc. Position RPM MOD 0 switch to down. GALVO SIM meter shall indicate 29 to 37 microamperes. If necessary, readjust DC INPUT CALIBRATE control to obtain specified value. If dc input drops, readjust to +0.40 ±0.01 vdc. Return RPM MOD 0 switch to center.

NOTE

Paragraph 4.7.3.8.4 to be performed for Type II (Proofing) Instrumentation Assembly only.

4.7.3.8.4 Adjust DC INPUT CALIBRATE control for +11.0 ±0.1 vdc indication on digital dc voltmeter. Position REC DISC switch to down. GALVO SIM meter shall indicate 36 to 45 microamperes. Return REC DISC switch to center.

4.7.3.8.5 Position GALVO SIM switch to 3. GALVO SIM meter shall indicate 35 to 43 microamperes. Adjust DC INPUT CALIBRATE control for -15.5 ±0.1 vdc indication on digital dc voltmeter. Position RANGE GATE switch to down. GALVO SIM meter shall indicate 30 to 38 microamperes.

4.7.3.8.6 Return RANGE RATE switch to center and position AMPL THRESH switch to down. GALVO SIM meter shall indicate 27 to 35 microamperes.

NOTE

Paragraph 4.7.3.8.7 to be performed for Type II (Proofing) Instrumentation Assembly only.

4.7.3.8.7 Return AMPL THRESH switch to center and position GALVO SIM switch to 4 and GALVO SIM polarity switch to REV. Momentarily hold ROLL DR DIFF switch to down. GALVO SIM meter shall indicate a slight upward transient when ROLL DR DIFF switch is switched to down position. Return GALVO SIM polarity switch to NORM. Return ROLL DR DIFF switch to center.

4.7.3.8.8 Position GALVO SIM switch to 5. Adjust DC INPUT CALIBRATE control for +33 ±0.1 vdc indication on digital dc voltmeter. Position +DOPPLER switch to down. GALVO SIM meter shall indicate 36 to 44 microamperes.

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4.7.3.8.9 Return +DOPPLER switch to center and adjust DC INPUT CALIBRATE control for +5.2 ±0.1 vdc indication on digital dc voltmeter. Position YAW COMMAND switch to L. GALVO SIM meter shall indicate 36 to 44 microamperes.

4.7.3.8.10 Position YAW COMMAND switch to R. GALVO SIM meter shall indicate 36 to 44 microamperes.

4.7.3.8.11 Return YAW COMMAND switch to center and position GALVO SIM switch to 6. Adjust DC INPUT CALIBRATE control for +33.0 ±0.1 vdc indication on digital voltmeter. Position -DOPPLER switch to down. GALVO SIM meter shall indicate 36 to 44 microamperes.

4.7.3.8.12 Return -DOPPLER switch to center and adjust DC INPUT CALIBRATE control for +5.2 ±0.1 vdc indication on digital dc voltmeter. Position PITCH COMMAND switch to DN. GALVO SIM meter shall indicate 36 to 44 microamperes.

4.7.3.8.13 Position PITCH COMMAND switch to UP. GALVO SIM meter shall indicate 36 to 44 microamperes.

4.7.3.8.14 Return PITCH COMMAND switch to center and adjust DC INPUT CALIBRATE control for +40.00 ±0.25 vdc indication on digital dc voltmeter. Position VALID ECHO INHIBIT switch to down. GALVO SIM meter shall indicate 29 to 37 microamperes.

4.7.3.8.15 Return VALID ECHO INHIBIT switch to center and position GALVO SIM switch to 7 and GALVO SIM polarity switch to REV. Position ARM switch to down. GALVO SIM meter shall indicate 10 to 16 microamperes.

4.7.3.8.16 Return ARM switch to center and position TYPE 1 FIRE-TYPE 2 FIRE switch to TYPE 2 FIRE and GALVO SIM polarity to NORM. GALVO SIM meter shall indicate a positive transient and decay to a steady state level of 8 to 10 microamperes.

4.7.3.8.17 Position TYPE 1 FIRE-TYPE 2 FIRE switch to TYPE 1 FIRE. GALVO SIM meter shall indicate a positive transient and decay to a steady state level of 3 to 5 microamperes. Return TYPE 1 FIRE-TYPE 2 FIRE switch to center.

NOTE

Paragraphs 4.7.3.8.18 and 4.7.3.8.19 are to be performed for Type II (Proofing) Instrumentation Assembly only.

4.7.3.8.18 Adjust DC INPUT CALIBRATE control to +1.0 ±0.1 vdc indication on digital dc voltmeter. Position CHAMBER switch to down. GALVO SIM meter shall indicate 22 to 30 microamperes.

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4.7.3.8.19 Return CHAMBER switch to center and adjust DC INPUT CALIBRATE control for +40.0 ±0.1 vdc indication on digital dc voltmeter. Position AGC THRESH switch to down. GALVO SIM meter shall indicate 15 to 21 microamperes. Return AGC THRESH switch to center.

4.7.3.8.20 Position GALVO SIM switch to 8 and GALVO SIM polarity switch to REV. GALVO SIM meter shall indicate 35 to 43 microamperes. Adjust DC INPUT CALIBRATE control for +15.5 ±0.1 vdc indication on digital dc voltmeter. Position TARGET < GATE switch to down. GALVO SIM meter shall indicate 26 to 34 microamperes.

4.7.3.8.21 Return TARGET < GATE switch to center. Position NOISE THRESH switch to down. GALVO SIM meter shall indicate 31 to 39 microamperes.

4.7.3.8.22 Return NOISE THRESH switch to center, GALVO SIM polarity switch to NORM, and position GALVO SIM switch to 9. Position EXERCISE HEAD PWR switch to off. Connect Patch Plug (NAVORD Drawing [illegible] 0565) to connector J4810. (See Figure 3.)

4.7.3.8.23 Position EXERCISE HEAD PWR switch to on. Momentarily depress ALD RESET switch on instrumentation assembly and observe that GALVO SIM meter indicates positive transient each time switch is depressed.

NOTE

Paragraphs 4.7.3.8.24 to 4.7.3.8.33 are to be performed for Type II (Proofing) Instrumentation Assembly only.

4.7.3.8.24 Adjust AC INPUT CALIBRATE control for 2.5 ±0.1 vrms (0° ∅) indication on AC VTVM. Position ROLL PEND switch to down. GALVO SIM meter shall indicate 26 to 34 microamperes. If no GALVO SIM meter indication, depress ALL RESET switch. Return ROLL PEND switch to center position.

4.7.3.8.25 Adjust AC INPUT CALIBRATE control for 2.5 ±0.1 vrms (180° ∅) on AC VTVM. Position PITCH PEND switch to down. GALVO SIM meter shall indicate 26 to 34 microamperes. If no GALVO SIM meter indication, depress ALD RESET switch.

4.7.3.8.26 Return PITCH PEND switch to center and position GALVO SIM switch to 11. Adjust DC INPUT CALIBRATE control for +5.2 ±0.1 vdc indication on digital DC voltmeter. Position PITCH ∠ DATA switch to down. GALVO SIM meter shall indicate 36 to 44 microamperes.

4.7.3.8.27 Return PITCH ∠ DATA switch to center and position FREQ IND switch to down. GALVO SIM meter shall indicate 36 to 44 microamperes.

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4.7.3.8.28 Return FREQ IND switch to center and position GALVO SIM switch to 12. Adjust AC INPUT CALIBRATE control for 2.0 ±0.1 vrms (180° ∅) indication on AC VTVM. Position YAW RATE switch to down. GALVO SIM meter shall indicate 36 to 44 microamperes.

4.7.3.8.29 Return YAW RATE switch to center. Momentarily hold YAW DR DIFF switch to down. GALVO SIM meter shall indicate a slight downward transient when YAW DR DIFF switch is switched to down position.

4.7.3.8.30 Adjust DC INPUT CALIBRATE control for +40.00 ±0.25 vdc. Return YAW DR DIFF switch to center and position GALVO SIM switch to 13. Position SAMPLE switch to down. GALVO SIM meter shall indicate 23 to 31 microamperes.

4.7.3.8.31 Return SAMPLE switch to center and adjust DC INPUT CALIBRATE control for +5.2 ±0.1 vdc indication on digital DC voltmeter. Position YAW DATA switch to down. GALVO SIM meter shall indicate 36 to 44 microamperes.

4.7.3.8.32 Return YAW DATA switch to center and position GALVO SIM switch to 14. Adjust AC INPUT CALIBRATE control for 2.0 ±0.1 vrms (180° ∅) indication on AC VTVM. Position PITCH RATE switch to down. GALVO SIM meter shall indicate 36 to 44 microamperes.

4.7.3.8.33 Return PITCH RATE switch to center. Momentarily hold PITCH DR DIFF switch to down. GALVO SIM meter shall indicate a slight downward transient when PITCH DR DIFF switch is switched to down position.

4.7.3.8.34 Reset AC INPUT CALIBRATE and DC INPUT CALIBRATE dials to approximately 500 dial setting and disconnect test equipment from test set.

4.7.4 Visual inspections

4.7.4.1 Identification marking. (Major)

4.7.4.1.1 Nomenclature, control callout, and identification data shall be correct, clear, and legible. (Reference: DL 2539153, DL 2539071, or DL 2539544.)

4.7.4.2 Interchangeability. (Major)

4.7.4.2.1 Connector receptacles shall be correct and capable of accepting mating unit without resorting to selection, modification, or fitting. (Reference: DL 2539153, DL 2539071, or DL 2539544.)

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4.7.4.2.2 Verify that location of four holes 0.375 inches minimum in diameter shall be in accordance with NAVORD Drawing 2197301 and shall be as follows:

(a)	Zone 15-E	2.843 ±0.030 inches
(b)	Zone 14-J	4.500 ±0.005 inches
(c)	Zone 14-J	9.000 ±0.005 inches
(d)	Zone H-12	0.719 ±0.030 inches
(e)	Zone H-11	2.810 ±0.005 inches
(f)	Zone G-11	6.437 ±0.005 inches
(g)	Zone D-14	3.875 ±0.005 inches
(h)	Zone D-14	7.750 ±0.005 inches

4.7.4.2.3 Diameter dimension shall be 11.330 inches maximum and located in zone 18-G of NAVORD Drawing 2197301.

4.7.4.2.4 Thickness dimension shall be 7.250 inches maximum and located in zone 11-N of NAVORD Drawing 2197301.

4.7.4.2.5 Verify that the battery assembly has satisfactorily met the requirements as specified in CD 2539219.

4.7.5 Process verification

4.7.5.1 Criteria stipulated in this paragraph and its subparagraphs are applicable to the production process instead of specific units of product. Inspections stipulated are general in scope and technique and frequency of verification will be contingent upon specific process conditions. Positive action by the contractor is required when process criteria findings indicate poor inspection by the contractor.

4.7.5.2 Lacing (Minor)

4.7.5.2.1 Lacing and cable configuration shall be in accordance with applicable drawings of DL 2539153, DL 2539071, or DL 2539544. Specific method of verifying adequacy of cable lacing shall include visual and dimensional verification.

4.7.5.3 Soldering (Major)

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4.7.5.3.1 Soldered joints shall be in accordance with good commercial practices. Unsoldered connections or those of a "cold" solder or "rosin" joint shall be considered defective.

4.7.5.4 Cleanliness (Major)

4.7.5.4.1 Presence of foreign material in assembly shall classify unit as defective. Foreign material includes anything not intended to be present.

4.7.6 Packaging, packing, and marking inspections

4.7.6.1 Packaging (Minor)

4.7.6.1.1 Level A packaging shall comply with DL 1330423. If level C packaging is used, packaging shall comply with commercial practice.

4.7.6.2 Packing (Major)

4.7.6.2.1 Level A packing shall be in accordance with MIL-P-116, Method III. If level C packing is used, the instrumentation assembly shall be protected against damage during direct shipment from supplier to first receiving activity for immediate use.

4.7.6.3 Marking (Major)

4.7.6.3.1 Destination and other shipment markings shall be correct, legible and in accordance with DL 1330423.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2c and 6.4).

5.1.1 Level A. Cleaning, drying, preservation, and packaging methods for level A shipment shall be in accordance with instructions contained in applicable documents listed in DL 1330423.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C as specified (see 6.2d and 6.4).

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5.2.1 Level A. Multiples of the instrumentation assembly, preserved and packaged in accordance with 5.1.1, shall be packed in accordance with MIL-P-116, Method III.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The instrumentation assembly, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal marking. In addition to markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with DL 1330423.

6. NOTES

6.1 Intended use and description

6.1.1 Intended use. The instrumentation assembly is for use in an Exercise Head Mark 85. It is of a Type I, Type II, or Type III configuration. Type I configuration is used in Exercise Head Mark 85 Mod 0 (Fleet). Type II configuration is used in Exercise Head Mark 85 Mod 1 (Proofing). The Type III configuration is used in Exercise Head Mark 85 Mod 2 (Fleet).

6.1.2 Description. Type I Instrumentation Assembly is for fleet use and provides the necessary interconnecting wires between the exercise head cables, Oscillograph and Logic assembly. Also provided are the necessary processing signal conditioning circuitry. Type II Instrumentation Assembly is for proofing use and is basically the same as Type I except for additional processing and signal conditioning circuitry. Type III Instrumentation Assembly is basically the same as Type I, except that Type III provides the means of monitoring either of two types of cource gyroscope inputs.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Type required (see 1.2).

(c) Required level of preservation and packaging.

(d) Required level of packing.

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(e) Required data.

(f) Assigned activity for first article and periodic tests.

(g) Any exceptions to this specification.

(h) All additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions. For the purpose of this specification, the following definitions have been established:

6.3.1 Unit condition

6.3.1.1 Nonoperating. The condition of the uninstalled instrumentation assembly when stored, transported, or handled (see 5.1.1).

6.3.1.2 Ready, nonoperating. The condition representative of the instrumentation assembly when assembled with end item, transported, and handled.

6.3.1.3 Ready. The condition representative of the instrumentation assembly when fully assembled with the end item and installed on the launching mechanism prior to operating.

6.3.1.4 Operating. The condition of the instrumentation assembly when performing its required function.

6.3.2 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (cps).

6.3.3 Shock. An abrupt change in the position of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's), of duration in units of time, milliseconds, (ms) and in the form of a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines tangent at maximum slope intersect the zero g line.

6.3.4 Acceleration (steady state). The time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second.

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6.3.5 Temperature. The temperature of the environment to which the instrumentation assembly may be exposed, expressed in degrees Fahrenheit (°F).

6.3.6 Pressure. The pressure of the environment to which the instrumentation assembly may be exposed, expressed in pounds per square inch absolute (psia).

6.3.7 Humidity. The relative humidity of the environment to which the instrumentation assembly is exposed, expressed as a percent.

6.3.8 Defective unit. A unit shall be considered defective if it does not meet all quality conformance examinations or tests specified, or if deviations are found through visual examination after environmental exposure.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A: This level shall be used for those items which are to be shipped to intermediate destinations or stored under indeterminate conditions for redistribution anywhere.

Level B: Not applicable.

Level C: This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.5 Storage life. The instrumentation assembly preserved and packaged in accordance with Section 5, is intended to withstand an unattended storage life of 5 years.

Preparing Activity
Navy - OS
Project No. 1355-N064

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

FOR

INSTRUMENTATION ASSEMBLY

DL 2539153 (Type I), DL 2539071 (Type II),

or DL 2539544 (Type III)

(For Exercise Head Mk 85 Mod 0, Mod 1, or Mod 2)

EQUIPMENT STATUS			
CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TEST AND EXAMINATION

INSTRUMENTATION ASSEMBLY DL 2539153 (Type I),

DL 2539071 (Type II), or DL 2539544 (Type III)

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the quality conformance tests and examinations of 4.7 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the instrumentation assembly bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The quality conformance tests and examinations of 4.7 shall be completed and the results recorded or verified in this report form prior to acceptance of the instrumentation assembly. Unless otherwise noted, the sequence of testing, recording and verification, as performed in 4.7, is mandatory.

20.2 Recording of results. Where space is provided to indicate the value of specific reading obtained, the specific reading shall be recorded if a limit or limits is given in the Limit column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limit column, the individual performing the test or inspection shall initial in the space provided in the Reading column, indicating that the requirement has been met satisfactorily.

30. QUALITY CONFORMANCE TEST AND EXAMINATION

		Reference	Limits	Reading
30.1	Configuration type	4.7	Type I/Type II/ Type III	_____
30.2	Test equipment	4.7.2	In calibration	_____

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	Reference	Limits	Reading
30.3 Test procedure			
30.3.1 Continuity tests	4.7.3.1		
From **To**			
J4806-A J4806-a		0.5 ohm max	_____
J4806-K J4805-V, T		0.5 ohm max	_____
J4806-K J4804-s, t		0.5 ohm max	_____
J4806-K J4810-X		0.5 ohm max	_____
J4806-S J4810-C (See NOTE 1)		0.5 ohm max	_____
J4806-T J4810-F		0.5 ohm max	_____
J4806-U J4810-E		0.5 ohm max	_____
J4806-X J4810-KK		0.5 ohm max	_____
J4806-Y J4810-JJ		0.5 ohm max	_____
J4806-b J4810-AA		0.5 ohm max	_____
J4806-c J4810-CC		0.5 ohm max	_____
J4806-c J4805-K		0.5 ohm max	_____
J4806-d J4805-H		0.5 ohm max	_____
J4806-e J4805-F		0.5 ohm max	_____
J4806-h J4805-W		0.5 ohm max	_____
J4806-j J4805-B		0.5 ohm max	_____
J4806-m J4805-L		0.5 ohm max	_____
J4806-u J4810-A		0.5 ohm max	_____

[1]NOTE: For Type I only.

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	Reference	Limits	Reading
30.3.1 (Cont'd)			
From To			
J4806-w J4810-B		0.5 ohm max	_____
J4806-x J4810-P		0.5 ohm max	_____
J4806-y J4805-M		0.5 ohm max	_____
J4806-AA J4805-Z		0.5 ohm max	_____
J4806-BB J4805-Y		0.5 ohm max	_____
J4806-CC J4810-N		0.5 ohm max	_____
J4806-DD J4810-R		0.5 ohm max	_____
J4806-EE J4810-H		0.5 ohm max	_____
J4806-EE J4804-Z		0.5 ohm max	_____
J4806-EE J4804-b		0.5 ohm max	_____
J4806-EE J4804-d		0.5 ohm max	_____
J4806-EE J4804-f		0.5 ohm max	_____
J4806-EE J4804-h		0.5 ohm max	_____
J4806-EE J4804-j		0.5 ohm max	_____
J4806-EE J4804-m		0.5 ohm max	_____
J4806-EE J4804-p		0.5 ohm max	_____
J4806-EE J4803-V		0.5 ohm max	_____
J4806-FF J4805-N		0.5 ohm max	_____
J4805-A J4810-MM		0.5 ohm max	_____

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Reference		Limits	Reading
30.3.1 (Cont'd)			
From	To		
J4805-C	P2302-A	0.5 ohm max	_____
J4805-D	J4810-EE	0.5 ohm max	_____
J4805-E	J4810-DD	0.5 ohm max	_____
J4805-J	J4810-FF	0.5 ohm max	_____
J4805-U	J4810-LL	0.5 ohm max	_____
J4805-AA	J4810-K	0.5 ohm max	_____
J4805-BB	J4810-S	0.5 ohm max	_____
J4805-CC	J4804-q	0.5 ohm max	_____
J4805-DD	J4804-r	0.5 ohm max	_____
J4804-s,t	TB4802, GND	0.5 ohm max	_____
J4803-L	TB4802, PING DRIVE	0.5 ohm max	_____
J4800-L	J4803-J	0.5 ohm max	_____
J4810-M	TB4802 STEERAWAY DISABLE	0.5 ohm max	_____
J4810-V	J4804-X	0.5 ohm max	_____
J4806-S (See NOTE 2)	TB4802, C. G. IN	0.5 ohm max	_____
J4810-J (See NOTE 2)	TB4802, HATS	0.5 ohm max	_____
J4810-C (See NOTE 2)	TB4802, NORM	0.5 ohm max	_____

[2]NOTE: For Type II and Type III only.

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			Reference	Limits	Reading
30.3.2	Type II continuity and resistance tests		4.7.3.2		
	From	To			
	P4808-A	P2302-M		0.5 ohm max	_____
	P4808-B	P2302-P		0.5 ohm max	_____
	P4808-C	P2302-N		0.5 ohm max	_____
	P4809-A	J4804-U		0.5 ohm max	_____
	P4809-B	J4804-V		0.5 ohm max	_____
	J4806-EE	J4804-K		0.5 ohm max	_____
	J4806-EE	J4804-M		0.5 ohm max	_____
	J4806-EE	J4804-R		0.5 ohm max	_____
	J4806-EE	J4804-S		0.5 ohm max	_____
	P4809-D	J4806-z	4.7.3.2.2	313 to 347 ohms	_____
30.3.3	Resistance tests		4.7.3.3		
	From	To			
30.3.3.1	J4804-i	J4810-S	4.7.3.3.1	256 k to 284 k ohms	_____
30.3.3.2	J4804-Y	J4810-K	4.7.3.3.2	256 k to 284 k ohms	_____
30.3.3.3	J4806-J	J4804-g	4.7.3.3.3	31.3 k to 34.7 k ohms	_____
30.3.3.4	J4810-D	J4804-k	4.7.3.3.4	950 to 1050 ohms	_____
30.3.3.5	J4810-[illegible]	J4804-n	4.7.3.3.5	2.56 k to 2.84 k ohms	_____

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		Reference	Limits	Reading
30.3.4	ALD/HLD RESET switch test	4.7.3.4		
	ALD/HLD RESET switch depressed	4.7.3.4.2	0.2 ohm max	_____
	ALD/HLD RESET switch released		500 k ohms min	
30.3.5	3-minute snake	4.7.3.6		
	EXERCISE HEAD PWR on	4.7.3.6.2	Minutes indication	_____
	3 MIN SNAKE indication	4.7.3.6.3	Minutes indication	_____
	Time interval	4.7.3.6.4	2.5 to 3.5 minutes	_____
30.3.6	Battery load test	4.7.3.7		
	DCO SQUIB and TERMINATION SQUIB indicators shall	4.7.3.7.1	illuminate	_____
	Panel meter indication	4.7.3.7.2	-2.5 vdc min	_____
	Panel meter indication	4.7.3.7.3	-2.5 vdc min	_____
	DCO SQUIB and TERMINATION SQUIB indicators shall	4.7.3.7.4	extinguish	_____
30.3.7	Signal conditioning circuits	4.7.3.8		
30.3.7.1	GALVO SIM position 1	4.7.3.8.2		

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	Reference	Limits	Reading
30.3.7.1 (Cont'd)			
VALID ECHO switch down		36 to 44 μa	_____
30.3.7.2 GALVO SIM position 2	4.7.3.8.3		
(a) RPM MOD 0 switch down		29 to 37 μa	_____
NOTE: Step (b) applies to Type II only.			
(b) REC DISC switch down	4.7.3.8.4	36 to 45 μa	_____
30.3.7.3 GALVO SIM position 3	4.7.3.8.5		
(a) +40 vdc		35 to 43 μa	_____
(b) RANGE GATE switch down		30 to 38 μa	_____
(c) AMPL THRESH switch down	4.7.3.8.6	27 to 35 μa	_____
NOTE: Step 30.3.7.4 applies to Type II (Proofing) only			
30.3.7.4 GALVO SIM position 4	4.7.3.8.7		
(a) ROLL DR DIFF switch down momentarily		slight upward transient	_____

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	Reference	Limits	Reading
30.3.7.5 GALVO SIM position 5	4.7.3.8.8		
(a) +DOPPLER switch down		36 to 44 μa	_____
(b) YAW COMMAND switch to L	4.7.3.8.9	36 to 44 μa	_____
(c) YAW COMMAND switch to R	4.7.3.8.10	36 to 44 μa	_____
30.3.7.6 GALVO SIM position 6	4.7.3.8.11		
(a) -DOPPLER switch down		36 to 44 μa	_____
(b) PITCH COMMAND switch to DN	4.7.3.8.12	36 to 44 μa	_____
(c) PITCH COMMAND switch to UP	4.7.3.8.13	36 to 44 μa	_____
(d) VALID ECHO INHIBIT switch	4.7.3.8.14	29 to 37 μa	_____
30.3.7.7 GALVO SIM position 7	4.7.3.8.15		
(a) ARM switch down		10 to 16 μa	_____
(b) TYPE 1 FIRE-TYPE 2 FIRE switch to TYPE 2 FIRE	4.7.3.8.16	positive transient decay to 8 to 10 μa	_____
(c) TYPE 1 FIRE-TYPE 2 FIRE switch to TYPE 1 FIRE	4.7.3.8.17	positive transient decay to 3 to 5 μa	_____

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	Reference	Limits	Reading
30.3.7.7 (Cont'd)			
NOTE Steps (d) and (e) apply to Type II only.			
(d) CHAMBER switch down	4.7.3.8.18	22 to 30 μa	_____
(e) AGC THRESH switch down	4.7.3.8.19	15 to 21 μa	_____
30.3.7.8 GALVO SIM position 8	4.7.3.8.20		
(a) -40 vdc		35 to 43 μa	_____
(b) TARGET ∠ GATE switch down		26 to 34 μa	_____
(c) NOISE THRESH switch down	4.7.3.8.21	31 to 39 μa	_____
30.3.7.9 GALVO SIM position 9	4.7.3.8.22		
Inst. timing marks	4.7.3.8.23	positive transient	_____
NOTE Paragraphs 30.3.7.10 through 30.3.7.14 apply to Type II (Proofing) only.			
30.3.7.10 GALVO SIM position 9	4.7.3.8.24		
(a) ROLL PEND switch down		26 to 34 μa	_____

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		Reference	Limits	Reading
30.3.7.10	(Cont'd)			
	(b) PITCH PEND switch down	4.7.3.8.25	26 to 34 μa	_____
30.3.7.11	GALVO SIM position 11	4.7.3.8.26		
	(a) PITCH ∠ DATA switch to down		36 to 44 μa	_____
	(b) FREQ IND switch down	4.7.3.8.27	36 to 44 μa	_____
30.3.7.12	GALVO SIM position 12	4.7.3.8.28		
	(a) YAW RATE switch down		36 to 44 μa	_____
	(b) YAW DR DIFF switch down momentarily	4.7.3.8.29	slight downward transient	_____
30.3.7.13	GALVO SIM position 13	4.7.3.8.30		
	(a) SAMPLE switch down		23 to 31 μa	_____
	(b) YAW DATA switch down	4.7.3.8.31	36 to 44 μa	_____
30.3.7.14	GALVO SIM position 14	4.7.3.8.32		
	(a) PITCH RATE switch down		36 to 44 μa	_____
	(b) PITCH DR DIFF switch down momentarily	4.7.3.8.33	slight downward transient	_____

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		Reference	Limits	Reading
30.4	Visual inspections			
30.4.1	Identification marking	4.7.4.1		
30.4.2	Interchangeability			
	Connectors	4.7.4.2.1		
	Holes (4)	4.7.4.2.2	0.375 inches min	_____
	Hole location			
	(a)		2.843 ±0.030 in.	_____
	(b)		4.500 ±0.005 in.	_____
	(c)		9.000 ±0.005 in.	_____
	(d)		0.719 ±0.030 in.	_____
	(e)		2.810 ±0.005 in.	_____
	(f)		6.437 ±0.005 in.	_____
	(g)		3.875 ±0.005 in.	_____
	(h)		7.750 ±0.005 in.	_____
	Diameter dimension	4.7.4.2.3	11.330 in. max	_____
	Thickness dimension	4.7.4.2.4	1.250 in. max	_____
	Battery assembly	4.7.4.2.5	satisfactory	_____
30.5	Process verification			
30.5.1	Lacing	4.7.5.2		_____

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		Reference	Limits	Reading
30.5.2	Soldering	4.7.5.3		
30.5.3	Cleanliness	4.7.5.4		
30.6	Packaging, packing, and marking inspection			
30.6.1	Packaging	4.7.6.1		
30.6.2	Packing	4.7.6.2		
30.6.3	Marking	4.7.6.3		

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-82431A(OS) INSTRUMENTATION ASSEMBLY (FOR EXERCISE HEAD MK 85 MOD 0, Mod 1 & Mod 2

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426
APR 62

C-0870

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